

$$\tilde{Z} = \int_{-\infty}^{\infty} dp e^{-\frac{1}{2}ap^2 + Jp}$$

Complete the square $-\frac{1}{2}ap^2 + Jp$

$$= -\frac{1}{2}\left(p^2 - 2\frac{J}{a}p + \left(\frac{J}{a}\right)^2 - \left(\frac{J}{a}\right)^2\right)$$

$$= -\frac{1}{2}a\left[\left(p - \frac{J}{a}\right)^2 - \left(\frac{J}{a}\right)^2\right]$$

$$= -\frac{1}{2}a\left(p - \frac{J}{a}\right)^2 + \frac{J^2}{2a}$$

$$\tilde{Z} = e^{J^2/2a} \int_{-\infty}^{\infty} dp e^{-\frac{1}{2}a(p-J/a)^2}$$

$$= e^{J^2/2a} \int_{-\infty}^{\infty} d\tilde{p} e^{-\frac{1}{2}a\tilde{p}^2}, \quad \tilde{p} = p - J/a$$

• Now use the $\mathbb{R}^2$ trick

$$\left(\int dp e^{-\frac{1}{2}p^2}\right)^2 = \int dx e^{-\frac{1}{2}x^2} \int dy e^{-\frac{1}{2}y^2}$$

$$= \int dx dy e^{-\frac{1}{2}(r^2)} \quad r^2 = x^2 + y^2$$

$$dx dy = r dr d\theta$$

$$= \left(\int_0^{2\pi} d\theta\right) \int_0^{\infty} dr r e^{-\frac{1}{2}r^2} = 2\pi \int_0^{\infty} dr r e^{-\frac{1}{2}r^2}$$

$$u = r^2 \Rightarrow du = 2r dr$$

$$= \pi \int_0^{\infty} du e^{-u/2} = -2\pi(0 - 1) = 2\pi$$

$$\therefore \int dp e^{-\frac{1}{2}p^2} = \sqrt{2\pi}$$

$$\int_{-\infty}^{\infty} dp e^{-ap^2} = \sqrt{\frac{\pi}{a}}$$

$$\therefore \int dp e^{-ap^2 + Jp} = \sqrt{\frac{\pi}{a}} e^{J^2/4a}$$

$$\langle q_{j+1} | e^{-iH\delta t} | q_j \rangle = e^{-iV(q_j, t_j)\delta t} \int \frac{dp}{2\pi} e^{-i\frac{p^2}{2m}\delta t} e^{ip(x_{j+1}-x_j)}$$

$$= e^{-iV(q_j, t_j)\delta t} \sqrt{\frac{\pi}{i\frac{\delta t}{2m}}} e^{i\frac{(x_{j+1}-x_j)^2}{2\delta t} m}$$

• In the limit $\delta t \rightarrow 0$

$$\dot{x}_j = \frac{x_{j+1} - x_j}{\delta t}$$

$$= e^{-iV(q_i, t_i)\delta t} / e^{i\frac{m}{2}\dot{q}_i^2 \delta t}$$

$$L = \frac{1}{2} m \dot{q}^2 - V(q_i, t_i)$$

$$\therefore \langle q_{j+1} | e^{-iH\delta t} | q_j \rangle = N e^{iL(q, \dot{q}) \delta t}$$

$$H(q, p) \rightarrow L(q, \dot{q}) \quad (\text{Legendre transform})$$

$$\begin{aligned} \langle q'' | q' \rangle &= N^n \int dq_1 \dots dq_n e^{iL(q_1, \dot{q}_1) \delta t} \dots e^{iL(q_n, \dot{q}_n) \delta t} \\ &= \lim_{\delta t \rightarrow 0} \tilde{N} \int dq_1 \dots dq_n e^{i \sum_i L(q_i, \dot{q}_i) \delta t} \\ &= \tilde{N} \int_{x(t_i)=x_i}^{x(t_f)=x_f} Dx(t) e^{iS[q]} \end{aligned}$$

N : infinite const.

$$\int Dx(t) = \lim_{N \rightarrow \infty} \left(\prod_{i=1}^N \int dq_i \right)$$

* Path int in QFT.

VEV: $\langle 0; t_f | 0; t_i \rangle$

Schrodinger Picture $\hat{\phi}(\vec{x})$

$$\hat{\phi}(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{ipx} + a_p^\dagger e^{-ipx})$$

$$\hat{\pi}(\vec{x}) = -i \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\omega_p}{2}} (a_p e^{i\vec{p}\vec{x}} + a_p^\dagger e^{-i\vec{p}\vec{x}})$$

$$[\hat{\phi}(\vec{x}), \hat{\pi}(\vec{y})] = i\delta^3(\vec{x} - \vec{y})$$

* Eigenstates of $\hat{\phi}$:

$$\left\{ \begin{aligned} \hat{\phi}(\vec{x}) | \Phi \rangle &= \Phi(\vec{x}) | \Phi \rangle \\ \hat{\pi} | \Pi \rangle &= \Pi(\vec{x}) | \Pi \rangle \end{aligned} \right.$$

"|x>"

"|p>"

$$\langle \pi | \Phi \rangle = \exp\left(-i \int d^3x \pi(\vec{x}) \Phi(\vec{x})\right)$$

$$\langle p | x \rangle = e^{ipx}$$

$$\begin{aligned}\langle \Phi' | \Phi \rangle &= \int D\pi \langle \Phi' | \pi \rangle \langle \pi | \Phi \rangle \\ &= \int D\pi e^{-i \int d^3x \pi(\vec{x}) (\Phi(\vec{x}) - \Phi'(\vec{x}))} \\ \text{" } \langle x | x' \rangle &= \delta(x - x')\end{aligned}$$

obtain

$$\langle 0; t_f | 0; t_i \rangle = N \int D\Phi(\vec{x}, t) e^{i S[\Phi]}$$

Generating functionals:

$$Z[J] = \int D\phi e^{i S[\phi] + i \int d^4x J(x) \phi(x)}$$

$$Z[0] = \int D\phi e^{i S[\phi]} = \langle \Omega | \Omega \rangle = 1$$

* Variational Partial derivative:

$$J(x) = \int d^4y \delta(x-y) J(y)$$

then

$$\frac{\delta J(x)}{\delta J(y)} = \delta^4(x-y)$$

$$J(x) = \int d^4y \frac{\delta J(x)}{\delta J(y)} J(y)$$

$$\text{or } \delta J(x) = \int d^4y \frac{\delta J(x)}{\delta J(y)} \delta J(y)$$

$$\frac{\delta}{\delta J(x)} \int d^4x J(x) \phi(x) = \phi(x)$$

•

$$Z[J] = \int D\phi e^{i \int d^4x (\mathcal{L} + J\phi)}$$

$$\delta Z[J] / \delta J(x) = \int D\phi i \phi(x) e^{i \int d^4x (\mathcal{L} + J\phi)}$$

$$\Rightarrow \left. \frac{\delta Z[J]}{i \delta J(x)} \right|_{J=0} = Z[0] \langle \Omega | T \{ \phi(x) \} | \Omega \rangle$$

$$\therefore \left(\frac{1}{i Z[0]} \right) \left. \frac{\delta Z[J]}{\delta J(x)} \right|_{J=0} = \langle \Omega | T \{ \phi(x) \} | \Omega \rangle$$

$$- \delta Z[J]$$

$$\frac{\delta}{\delta J(x)} \frac{\delta Z[J]}{\delta J(x)} = \dots$$

$$(i\delta J(x,1) \dots (i\delta J(x_n)) \Big|_{J=0} \frac{1}{Z[0]} = \langle \Omega | T | \phi_1 \dots \phi_n | \Omega \rangle$$

Free field theory: $\mathcal{L} = -\frac{1}{2} \phi (\square + m^2)$

$$\tilde{\varphi}(k) = \int d^4x \, e^{-ikx} \varphi(x) \quad , \quad \varphi(x) = \int \frac{d^4k}{(2\pi)^4} e^{ikx} \tilde{\varphi}(k)$$

$$kx = -k^0 t + \vec{k} \cdot \vec{x}$$

$$\bullet i \int d^4x [\mathcal{L}_0 + \mathcal{J}\varphi] \\ = i \int d^4x \left(-\frac{1}{2} \phi(x) (\square + m^2) \phi(x) + \mathcal{J}(x) \phi(x) \right)$$

$$\textcircled{a} -\frac{1}{2} \int d^4x \int \frac{d^4k}{(2\pi)^4} e^{ikx} \tilde{\varphi}(k) (-\square + m^2) \int \frac{d^4\ell}{(2\pi)^4} e^{i\ell x} \tilde{\varphi}(\ell) \\ = -\frac{1}{2} \int \frac{d^4k d^4\ell}{(2\pi)^8} \tilde{\varphi}(k) \tilde{\varphi}(\ell) (\ell^2 + m^2) \int d^4x e^{ix(k+\ell)} \\ \leadsto (2\pi)^4 \delta(k+\ell) \quad k = -\ell$$

$$= -\frac{i}{2} \int \frac{d^4k}{(2\pi)^4} \tilde{\varphi}(k) \tilde{\varphi}(-k) (k^2 + m^2)$$

$$\textcircled{b} = \frac{i}{2} \int d^4x [\mathcal{J}(x) \phi(x) + \mathcal{J}(x) \phi(x)] \\ = \frac{i}{2} \int \frac{d^4k d^4\ell}{(2\pi)^8} e^{ix(k+\ell)} [\tilde{\mathcal{J}}(k) \tilde{\varphi}(\ell) + \tilde{\mathcal{J}}(\ell) \tilde{\varphi}(k)] \\ = \frac{i}{2} \int \frac{d^4k}{(2\pi)^4} [\tilde{\mathcal{J}}(k) \tilde{\varphi}(-k) + \tilde{\mathcal{J}}(-k) \tilde{\varphi}(k)]$$

$$\therefore i \int d^4x (\mathcal{L} + \mathcal{J}\varphi) = \frac{i}{2} \int \frac{d^4k}{(2\pi)^4} \left[-\tilde{\varphi}(k) (k^2 + m^2) \tilde{\varphi}(-k) \right. \\ \left. + \tilde{\mathcal{J}}(k) \tilde{\varphi}(-k) + \tilde{\mathcal{J}}(-k) \tilde{\varphi}(k) \right]$$

* We now make a change of variables

$$= (k^2 + m^2) \tilde{\varphi} \left(\tilde{\varphi} - \frac{\tilde{\mathcal{J}}(k)}{k^2 + m^2} \right)$$

$$\text{so} \quad \tilde{\chi}(k) \equiv \tilde{\varphi}(k) - \frac{\tilde{\mathcal{J}}(k)}{k^2 + m^2}$$

$$= \frac{i}{2} \int \frac{d^4k}{(2\pi)^4} \left[-(\tilde{\chi} + \tilde{\mathcal{J}}) (k^2 + m^2) (\chi + \tilde{\mathcal{J}}) + \tilde{\mathcal{J}}(\chi + \tilde{\mathcal{J}}) + \right. \\ \left. = \frac{i}{2} \int \frac{d^4k}{(2\pi)^4} \left[-\tilde{\chi}(-k) (k^2 + m^2) \chi(k) - \frac{\tilde{\mathcal{J}}(-k) \tilde{\mathcal{J}}(k)}{(k^2 + m^2)} - \tilde{\mathcal{J}}(-k) \tilde{\chi}(k) \right. \right. \\ \left. \left. - \tilde{\mathcal{J}}(k) \tilde{\chi}(-k) + \tilde{\mathcal{J}}(k) \tilde{\mathcal{J}}(-k) \right] \right]$$

$$+ \tilde{J}(-k) \tilde{X}(k) + \tilde{J}(-k) \tilde{J}(k) / (k^2 + m^2)$$

$$= \frac{i}{2} \int \frac{d^4 k}{(2\pi)^4} \left(\frac{\tilde{J}(k) \tilde{J}(-k)}{k^2 + m^2} - \tilde{X}(-k) (k^2 + m^2) \tilde{X}(k) \right)$$

$$Z_0[J] = \int \mathcal{D}X e^{i/2 \int \frac{d^4 k}{(2\pi)^4} \frac{\tilde{J}^2}{k^2 + m^2} - i/2 \int \frac{d^4 k}{(2\pi)^4} \tilde{X}(-k) (k^2 + m^2) \tilde{X}(k)}$$

$$= e^{i/2 \int \frac{d^4 k}{(2\pi)^4} \frac{\tilde{J}^2}{k^2 + m^2}}$$

Now, take $J=0$

$$Z_0[0] = \int \mathcal{D}X e^{-i/2 \int d^4 k X} = \langle 0|0 \rangle_{J=0} = 1$$

Thus,

$$Z_0[J] = \exp \left[\frac{i}{2} \int \frac{d^4 k}{(2\pi)^4} \frac{\tilde{J}(k) \tilde{J}(-k)}{k^2 + m^2} \right]$$

$$\int \frac{d^4 k}{(2\pi)^4} \int d^4 x \int d^4 y e^{-ikx} e^{iky} \left(\frac{1}{k^2 + m^2} \right) J(x) J(y)$$

$$= \int d^4 y d^4 x J(x) \Delta(x-y) J(y)$$

where $\Delta(x-y) = \int \frac{d^4 k}{(2\pi)^4} \frac{e^{ik(x-y)}}{k^2 + m^2}$

$$Z_0[J] = \exp \left[\frac{i}{2} \int d^4 x d^4 x' J(x) \Delta(x-x') J(x') \right]$$

$$(-\partial_x^2 + m^2) \Delta(x-x') = \delta^4(x-x')$$

4pt fun

$$Z_0[J] = e^{\frac{i}{2} \int dx \Delta_{xy} J_y}$$

$$\langle \Omega | T \phi_1 \phi_2 \phi_3 \phi_4 | \Omega \rangle = \left(\frac{1}{Z_0[0]} \right) \frac{\partial^4 Z_0[J]}{i\delta J_1 i\delta J_2 i\delta J_3 i\delta J_4}$$

$$= \frac{1}{Z_0[0]} \left(\frac{\partial}{i\delta J} \right)^3 \frac{\partial}{i\delta J_4} e^{i/2 \int dx \Delta_{xy} J_y}$$

$$= \frac{1}{Z_0[0]} \left(\frac{\partial}{i\delta J} \right)^3 \left(\frac{1}{i} \right) \left(\frac{1}{2} \right) (\delta_{ux} \Delta_{xy} J_y + J_x \Delta_{xy} \delta_{y4}) Z$$

$$= \frac{1}{2} \frac{1}{Z_0[0]} \left(\frac{\partial}{i\delta J} \right)^2 \frac{\partial}{\delta J_3} () Z$$

$$= \frac{1}{2} \left(\frac{1}{Z_0[0]} \right) \left(\frac{\partial}{i\delta J} \right)^2 (\Delta_{uy} J_y + J_x \Delta_{xu}) (\Delta_{3u} \delta_u + J_z \Delta_{z3}) Z$$

$$= \frac{1}{2} \left(\frac{1}{Z_0[0]} \right) (\Delta_{u3} + \Delta_{3u}) Z$$

$$\begin{aligned}
 &= (1/2) \text{Tr} \left(\frac{1}{i} \Delta_{42} \Delta_{32} \Delta_{22} \Delta_{12} \Delta_{11} \Delta_{41} \Delta_{31} \Delta_{21} \Delta_{11} \right) \\
 &= (1/2) \text{Tr} \left(\frac{1}{i} \Delta_{42} \Delta_{32} \Delta_{22} \Delta_{12} \Delta_{11} \Delta_{41} \Delta_{31} \Delta_{21} \Delta_{11} \right) \\
 &\quad + \Delta_{41} \Delta_{31} \Delta_{21} \Delta_{11} \Delta_{42} \Delta_{32} \Delta_{22} \Delta_{12} \Delta_{11} \\
 &\quad + 2 \Delta_{34} \Delta_{23} \Delta_{12} \Delta_{11} \Delta_{41} \Delta_{31} \Delta_{21} \Delta_{11} \\
 &\quad + 2 \Delta_{34} \Delta_{23} \Delta_{12} \Delta_{11} \Delta_{41} \Delta_{31} \Delta_{21} \Delta_{11}
 \end{aligned}$$

$$= \Delta_{34} \Delta_{12} + \Delta_{23} \Delta_{14} + \Delta_{13} \Delta_{24}$$

$$\langle 0 | T \{ \phi_0(x_1) \dots \phi_0(x_n) \} | 0 \rangle = \text{diagram 1} + \text{diagram 2} + \dots$$

$$\langle \Omega | T \{ \phi_1 \dots \phi_n \} | \Omega \rangle = \frac{\langle 0 | T \{ \phi_0 \dots \phi_n e^{i \int \mathcal{L}_I} \} | 0 \rangle}{\langle 0 | T \{ e^{i \int \mathcal{L}_I} \} | 0 \rangle}$$

$$\langle \Omega | T \{ \phi_1 \dots \phi_n \} | \Omega \rangle = \frac{1}{Z[\mathcal{J}]} \int D\phi \mathcal{L}(\phi) \phi_1 \dots \phi_n e^{i S[\phi]}$$

$$Z[\mathcal{J}] \equiv \langle \Omega | \Omega \rangle_{\mathcal{J}} = \int D\phi e^{i \int d^4x [\mathcal{L}_0 + \mathcal{L}_1 + \mathcal{J}\phi]}$$

$$\begin{aligned}
 Z[\mathcal{J}] &= \int D\phi e^{i \int d^4x \mathcal{L}_0 + \mathcal{J}\phi} \left(\sum_{n=0}^{\infty} \frac{(i \int d^4x \mathcal{L}_1)^n}{n!} \right) \\
 &= \int D\phi \sum_{n=0}^{\infty} \frac{(i \int d^4x \mathcal{L}_1[\phi])^n}{n!} e^{i \int d^4x (\mathcal{L}_0 + \mathcal{J}\phi)}
 \end{aligned}$$

However Note that

$$\int D\phi F(\phi) e^{i \mathcal{J}\phi} = \int D\phi F\left(\frac{\delta}{i \delta \mathcal{J}}\right) e^{i \mathcal{J}\phi}$$

ex: $F = \phi, \phi^2$

$$\int D\phi \phi e^{i \mathcal{J}\phi} = \int D\phi \frac{\delta}{i \delta \mathcal{J}} e^{i \mathcal{J}\phi}$$

$$Z[\mathcal{J}] = \int D\phi \sum_{n=0}^{\infty} \frac{(i \int d^4x \mathcal{L}_1[\frac{\delta}{i \delta \mathcal{J}}])^n}{n!} e^{i \int (\dots)}$$

$$\begin{aligned}
 &= e^{i \int d^4x \mathcal{L}_1 \left[\frac{\delta}{i \delta J(x)} \right]} \int D\varphi e^{i \int d^4x (\mathcal{L}_0 + J\varphi)} \\
 &= e^{i \int d^4x \mathcal{L}_1 \left(\frac{\delta}{i \delta J(x)} \right)} Z_0[J]
 \end{aligned}$$

$$Z[J] = e^{i \int d^4x \mathcal{L}_1 \left(\frac{\delta}{i \delta J(x)} \right)} Z_0[J]$$

* For

$$\mathcal{L} = -\frac{1}{2} Z_\varphi \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} Z_m m^2 \varphi^2 + \frac{1}{6} Z_g g \varphi^3 + Y \varphi$$

$$\mathcal{L}_0 = -\frac{1}{2} \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} m^2 \varphi^2$$

$$\text{for } \frac{i}{2} J_x \Delta x_y J_y$$

$$Z_0[J] = e$$

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1$$

$$\mathcal{L}_1 = \frac{1}{6} Z_g g \varphi^3 + \mathcal{L}_{ct}$$

$$\mathcal{L}_{ct} = -\frac{1}{2} (Z_\varphi - 1) \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} (Z_m - 1) m^2 \varphi^2 + Y \varphi$$

$$* \text{ Impose } Z[0] = \int D\varphi e^{i \int \mathcal{L}_0 + \mathcal{L}_1} = 1$$